Feature	Studies	Screens	Dir. cons.	I^2	q
Negative formal charge count	1	1	100%	0%	<0.001
Phosphorus count	2	3	100%	65%	0.118
Halogen count	1	1	100%	0%	0.002
Hall-Kier alpha	41	52	60%	86%	0.003
Aromatic heterocycles	**17**	**23**	**74%**	**73%**	**0.005**
Heteroatom fraction	41	52	63%	86%	0.009
Heteroatom:carbon ratio	41	52	63%	86%	0.009
Nitrogen fraction	**41**	**52**	**69%**	**86%**	**0.014**
TPSA per heavy atom	41	52	58%	85%	0.159
TPSA per MW	41	52	54%	85%	0.173
Oxygen fraction	41	52	56%	84%	0.253
Aromatic rings	25	33	70%	86%	0.388
Aromatic rings (RDKit)	25	33	70%	86%	0.388
N:O ratio	41	52	50%	79%	0.253
Saturated rings	36	45	58%	78%	0.641
Rings	37	46	61%	87%	0.768
Heterocycles	36	44	61%	85%	0.768
Aliphatic rings	36	45	44%	81%	0.948
Saturated heterocycles	36	44	45%	76%	0.768
Fraction sp3 carbons	40	51	47%	77%	0.768
Aliphatic heterocycles	36	44	45%	83%	0.507
H-bond donors	38	49	41%	92%	0.462
Amide bonds	14	16	50%	93%	0.487
Balaban J	41	52	63%	87%	0.040
logP per 100 MW	41	52	60%	88%	0.049
Carbon fraction	41	52	63%	86%	0.009
Oxygen count	41	52	58%	90%	0.002
Nitrogen count	36	46	61%	91%	0.002
Nitrogen count (RDKit)	36	46	61%	91%	0.002
Rotatable bond fraction	**41**	**52**	**67%**	**85%**	**<0.001**
Double bonds	40	51	63%	89%	<0.001
Heteroatom count	41	52	62%	91%	<0.001
TPSA	41	52	63%	92%	0.001
Sulfur count	15	20	60%	88%	0.015
H-bond acceptors	41	51	63%	91%	<0.001
Bertz complexity	**41**	**52**	**67%**	**89%**	**<0.001**
Exact molecular weight	**41**	**52**	**77%**	**88%**	**<0.001**
Heavy atoms	**41**	**52**	**77%**	**88%**	**<0.001**
Molecular weight	**41**	**52**	**77%**	**88%**	**<0.001**
Chi0	**41**	**52**	**77%**	**88%**	**<0.001**
Chi1	**41**	**52**	**77%**	**88%**	**<0.001**
Labute ASA	**41**	**52**	**77%**	**88%**	**<0.001**
Molar refractivity	**41**	**52**	**77%**	**88%**	**<0.001**
Carbon count	**41**	**52**	**77%**	**87%**	**<0.001**
Chi0n	**41**	**52**	**77%**	**88%**	**<0.001**
Chi1n	**41**	**52**	**79%**	**87%**	**<0.001**
VDW molecular volume	**41**	**52**	**77%**	**88%**	**<0.001**
Kappa 1	**41**	**52**	**77%**	**88%**	**<0.001**
sp3 carbons	**41**	**52**	**77%**	**87%**	**<0.001**
logP	**41**	**52**	**67%**	**87%**	**<0.001**
Kappa 3	**41**	**52**	**77%**	**88%**	**<0.001**
Rotatable bonds	**41**	**52**	**83%**	**88%**	**<0.001**
Kappa 2	**41**	**52**	**79%**	**87%**	**<0.001**
Formal charge	1	1	100%	0%	<0.001

−0.4 −0.3 −0.2 −0.1 0.0 0.1 0.2 0.3

Pooled effect ± 95% CI

Robust, FDR < 0.05 & ≥67% directional consistency

FDR < 0.05

FDR ≥ 0.05